

Newton's Laws of Motion

Newton's laws describe how forces affect the motion of objects in classical mechanics.

Key facts:

The first law says an object remains at rest or in uniform straight-line motion unless acted on by a net external force.

The second law states that net force equals mass times acceleration: $F = m \times a$.

The third law states that for every action force there is an equal and opposite reaction force.

Mass measures inertia, while weight is the gravitational force acting on mass.

Friction is a contact force that opposes relative motion or the tendency of motion between surfaces.

Momentum equals mass times velocity and is conserved in isolated systems.

Newton's laws work very well for everyday speeds and sizes but are extended by relativity and quantum mechanics in extreme cases.

Answerable questions:

Q: What does inertia mean? A: Inertia is the tendency of an object to resist changes in its state of motion.

Q: What is Newton's second law? A: It states that net force equals mass times acceleration, written as $F = m \times a$.